

QuantumDynamics Framework and Capability Notes

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Abstract

QuantumDynamics is a family of one-, two-, and three-coordinate, two-diabatic-state wavepacket and reduced-density-matrix tools for hydrogen-atom transfer. It combines split-operator propagation, optional Markovian or exponentially correlated Langevin centroid damping, optional stochastic modulation of either well, MPI-distributed trajectories, OpenMP/FFTW acceleration, and physical-unit input/output. It also provides quantum-detailed-balance density dynamics, strong-coupling non-Markovian HEOM, an exact finite spin-bath model, MPI-IO single-file output, exact restart, convergence campaigns, and controlled mechanism attribution. This note records equations, controls, tested behavior, analysis capability, and method boundaries.

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1 Scope and current status

Current code supports reproducible pure-state trajectories, ensemble kinetics, two- and three-coordinate wavepacket models, and finite-grid reduced density matrices under weak-coupling Davies or strong-coupling HEOM dynamics. Input validation, unit conversion, FFT ordering, random-stream portability, restart identity, serial and parallel HDF5, state-selective damping, and output safety have regression coverage. It remains a parameterized nuclear-dynamics model, not a first-principles electronic-structure engine.

2 Dynamical model

2.1 Two-state Hamiltonian

Wavefunction is

$$\Psi(x, t) = \begin{pmatrix} \psi_1(x, t) \\ \psi_2(x, t) \end{pmatrix}, \quad (1)$$

with Hamiltonian

$$\hat{H} = -\frac{1}{2m} \frac{\partial^2}{\partial x^2} \mathbb{1}_{2 \times 2} + \begin{pmatrix} V_{11}(x, t) & V_{12}(x) \\ V_{12}(x) & V_{22}(x, t) \end{pmatrix}. \quad (2)$$

Base diabatic wells are

$$V_{11}(x) = \frac{1}{2} k_1 (x - x_1)^2 + c_{4,1} (x - x_1)^4 + \Delta_1, \quad (3)$$

$$V_{22}(x) = \frac{1}{2} k_2 (x - x_2)^2 + c_{4,2} (x - x_2)^4 + \Delta_2. \quad (4)$$

Quartic terms are enabled by anharmonic model selections. Gaussian coupling is

$$V_{12}(x) = V_c \exp \left[-\frac{(x - x_c)^2}{2\sigma_c^2} \right], \quad (5)$$

with an exponential alternative. Lower and upper adiabatic surfaces are the eigenvalues of the local 2×2 diabatic potential matrix.

2.2 Split propagation

One Hamiltonian step uses symmetric potential splitting,

$$e^{-i\hat{H}\Delta t} \approx e^{-i\hat{V}\Delta t/2} e^{-i\hat{T}\Delta t} e^{-i\hat{V}\Delta t/2}. \quad (6)$$

Local potential exponentials use an analytic Hermitian 2×2 form. Kinetic propagation uses forward and inverse FFTW transforms. Dedicated plans are created for both diabatic components. Even and odd grid sizes have tested FFT frequency ordering.

2.3 Momentum Langevin bath

Markovian damping uses an exact finite-step Ornstein–Uhlenbeck centroid update consistent with classical fluctuation–dissipation statistics. Colored damping uses an exponential memory kernel,

$$\gamma(t) = \frac{\gamma}{\tau_c} e^{-t/\tau_c}, \quad \tau_c = \frac{2}{\text{FWHM}}, \quad (7)$$

with auxiliary memory and stationary random-force variables. The controls `damp_reactant` and `damp_product` select the diabatic component that receives the momentum translation. Enabling both recovers shared-centroid damping.

This thermostat is a classical-FDT pure-state construction. The QGLE label does not imply a full quantum-frequency-dependent fluctuation–dissipation kernel.

2.4 Quantum-FDT density-matrix path

`quantum_fdt_density.py` builds the full finite-grid system Hamiltonian, diagonalizes it, and propagates the reduced density matrix with a completely positive secular Davies generator. For each positive Bohr frequency ω , Drude–Ohmic emission and absorption rates are

$$\Gamma_{n \rightarrow m} = gJ(\omega)[n_B(\omega) + 1]|Q_{mn}|^2, \quad (8)$$

$$\Gamma_{m \rightarrow n} = gJ(\omega)n_B(\omega)|Q_{mn}|^2, \quad (9)$$

so that

$$\frac{\Gamma_{m \rightarrow n}}{\Gamma_{n \rightarrow m}} = e^{-\beta\omega}. \quad (10)$$

This KMS relation enforces quantum detailed balance at every resolved transition and makes the Gibbs density matrix stationary. Stored output includes the full complex density matrix, physical coordinate densities, populations, energies, and a KMS residual. This is the controlled weak-coupling, Markovian quantum-FDT limit. Strong-coupling non-Markovian quantum baths require HEOM or an equivalent influence-functional method and are outside this solver’s declared regime. They are handled by the separate HEOM path below.

2.5 Strong-coupling non-Markovian HEOM

`heom_nonmarkovian.py` treats a harmonic Gaussian environment with linear system coupling operator Q and Drude–Lorentz spectral density

$$J(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \quad (11)$$

where λ is reorganization energy and γ^{-1} is bath memory time. The finite-temperature bath correlation is decomposed as

$$C(t) = \sum_{k=0}^K c_k e^{-\nu_k t} + 2\Delta_K \delta(t), \quad (12)$$

with one Drude pole, K Matsubara poles, and residual white-tail correction Δ_K . Scaled auxiliary density operators $\tilde{\rho}_n$ obey

$$\begin{aligned} \dot{\tilde{\rho}}_n = & - \left(i\mathcal{L} + \sum_k n_k \nu_k + \Delta_K Q^\times Q^\times \right) \tilde{\rho}_n \\ & - i \sum_k \sqrt{(n_k + 1)|c_k|} Q^\times \tilde{\rho}_{n+e_k} \\ & - i \sum_k \sqrt{\frac{n_k}{|c_k|}} (c_k Q \tilde{\rho}_{n-e_k} - c_k^* \tilde{\rho}_{n-e_k} Q). \end{aligned} \quad (13)$$

The root ADO is the physical reduced density matrix; higher ADOs retain bath memory and system–bath correlations. No Born, Markov, or secular approximation is used. Hierarchy depth, Matsubara count, and retained system states are explicit convergence axes. The declared regime assumes a harmonic Gaussian

bath, linear coupling, and Drude spectral density. This follows the nonperturbative reduced-hierarchy construction of Ishizaki and Tanimura ([JPSJ 74, 3131 \(2005\)](#)) and scaled hierarchy stabilization of Shi *et al.* ([JCP 130, 084105 \(2009\)](#)).

Factorized, correlated-equilibrium, and correlated-projected preparations are available. Correlated equilibrium is obtained by solving the sparse stationary truncated HEOM with a unit-trace constraint; a system preparation operator is then applied to every ADO for correlated nonequilibrium initialization. At converged hierarchy and bath expansion, the stationary real-time hierarchy corresponds to the thermal hierarchy generated along imaginary time, as established by Tanimura ([JCP 141, 044114 \(2014\)](#)).

A 100 fs strong-coupling validation used $\lambda = 500 \text{ cm}^{-1}$, $\gamma^{-1} = 25 \text{ fs}$, ten retained system states, two Matsubara terms, and hierarchy depth four (35 ADOs). The reorganization/median-gap ratio was 4.47, maximum ADO norm was 0.966, trace and Hermiticity errors were below 5×10^{-16} , and the minimum root-density eigenvalue was -1.3×10^{-16} . Depth, bath-pole, and system-basis variants passed a 0.02 population tolerance; the largest population change was 1.70×10^{-4} .

A separate 10 K validation used eight Matsubara terms, hierarchy depth three, eight retained system states, and 220 ADOs. Depth four, a ninth Matsubara pole, and two extra system states all passed the same tolerance; the largest population change was 3.05×10^{-4} . Low-temperature calculations therefore remain supported, but require additional bath poles and convergence checks.

2.6 Exact non-Gaussian spin bath

`spin_bath_non_gaussian.py` exactly diagonalizes a two-state transfer system coupled to finite spin-1/2 modes,

$$H = H_S + \sum_j \frac{\omega_j}{2} \sigma_j^z + Q \sum_j g_j \sigma_j^x. \quad (14)$$

Thermal product-bath initialization, exact unitary evolution, and exact partial trace yield the reduced density matrix without Gaussian, Born, Markov, or secular assumptions. Output includes trace distance and information backflow. This finite non-Gaussian reference model follows exact spin-star treatments such as Semin, Sinayskiy, and Petruccione ([PRA 89, 012107 \(2014\)](#)). Hilbert-space cost is exponential, so the implementation enforces at most ten bath spins; it does not represent an arbitrary macroscopic non-Gaussian environment.

2.7 Stochastic potential bath

Either diabatic well can receive coordinate-shift, curvature, or enthalpy modulation. Noise can be white or Ornstein–Uhlenbeck colored and can be shared or independent between wells. Colored and white processes begin with their stationary Gaussian marginal distribution. Mode-dependent input amplitudes use length, force-constant, or energy units as appropriate.

2.8 Absorbing boundaries

Optional smooth edge masks suppress periodic FFT wraparound. Within an edge layer of width w , each amplitude is multiplied after a full time step by

$$M(x) = \exp \left[-\Gamma_{\text{abs}} \Delta t \left(\frac{d_{\text{edge}}(x)}{w} \right)^p \right]. \quad (15)$$

Absorbed norm represents outgoing flux. Norm-conservation tests apply only when edge absorption is disabled.

2.9 Two-coordinate propagation

`multidimensional_transfer.py` propagates $\Psi(x, y, t) = (\psi_1, \psi_2)^T$ with a two-dimensional FFT split operator. Coordinate x is the H-transfer coordinate and y is a promoting or bath mode with independent mass. Shifted harmonic valleys contain reaction-path coupling, and the diabatic coupling is Gaussian in both coordinates. The solver writes physical-unit two-dimensional densities and potential surfaces to HDF5.

2.10 Three-coordinate propagation and external PES data

`multidimensional_transfer_3d.py` propagates $\Psi(x, y, z, t)$ using a three-dimensional FFT split operator. Coordinate x is H transfer; y and z are independent promoting modes with separate masses and reaction-path couplings. The analytic model or an external Hermitian diabatic PES may supply V_{11} , V_{22} , and complex V_{12} .

`potential_data.py` defines the external HDF5 schema, rejects missing or incorrect units, validates finite monotone axes and array shapes, performs trilinear interpolation, and converts angstrom/cm⁻¹ data to atomic units once. Thus first-principles PES data can enter dynamics without manual unit handling. Electronic-structure generation remains the responsibility of an external code; no dynamics package can infer a first-principles PES from absent electronic-structure data.

3 Units and input boundary

Namelist input uses chemistry-facing units and converts once to atomic units. Output writers convert back to physical units.

Quantity	Input/output unit	Internal unit
Coordinate, widths	angstrom	bohr
Time	fs	atomic time
Energy and coupling	cm ⁻¹	hartree
Force constant	cm ⁻¹ angstrom ⁻²	hartree bohr ⁻²
Quartic coefficient	cm ⁻¹ angstrom ⁻⁴	hartree bohr ⁻⁴
Rates and FWHM	fs ⁻¹	atomic-time ⁻¹
Mass	u/amu	electron mass
Temperature	K	inverse hartree through β
Wave number	angstrom ⁻¹	bohr ⁻¹

4 Primary controls

Control	Meaning
<code>nx, xmin, xmax</code>	Periodic real-space grid and physical box.
<code>dt, nsteps, save_every</code>	Physical time step, total steps, and output cadence.
<code>ntraj, seed0</code>	Number of independent stochastic trajectories and base seed.
<code>temperature_k, mass_amu</code>	Thermodynamic temperature and transferred-particle mass.
<code>gamma, use_colored, kernel, fwhm</code>	Momentum bath strength and memory kernel.
<code>damp_reactant, damp_product</code>	Reactant/product diabatic damping switches.
<code>pot_model, k1, k2, x1, x2</code>	Well model, stiffnesses, and minima.
<code>v1_shift, v2_shift, c4_1, c4_2</code>	Energy offsets and quartic coefficients.
<code>v12, sigma, want_coupling</code>	Diabatic coupling magnitude, width, and switch.
<code>x0, p0, sigma0</code>	Initial Gaussian packet center, wave number, and width.
<code>bath_pot_*</code>	Stochastic potential mode, amplitude, color, and well selection.
<code>use_absorber</code>	Enable smooth edge absorption.
<code>absorber_width</code>	Edge-layer width in angstrom.
<code>absorber_rate</code>	Maximum mask rate in fs^{-1} .
<code>absorber_power</code>	Polynomial smoothness exponent.
<code>hdf5, out_prefix</code>	Output format and filename prefix.
<code>write_initial</code>	Write exact $t = 0$ snapshot before propagation.
<code>checkpoint_every</code>	Atomic, schema-versioned checkpoint cadence in steps.
<code>restart_from_checkpoint</code>	Continue from exact trajectory checkpoint.
<code>parallel_hdf5</code>	Write trajectory/frame hyperslabs to one MPI-IO HDF5 file.

Input may be supplied as `./qle_1d INPUT.nml`. Running `./qle_1d -help` cannot start a trajectory or overwrite output.

5 Parallel execution and reproducibility

MPI distributes complete trajectories; OpenMP and threaded FFTW accelerate each trajectory. Random streams depend on trajectory number rather than MPI rank. Default output uses independent trajectory files. With Parallel HDF5 enabled, ranks advance synchronized trajectory batches and write nonoverlapping trajectory-frame-grid hyperslabs into one `*.ensemble.h5` file through the HDF5 MPI-IO driver. A two-rank, three-trajectory regression checks layout, physical units, finite values, and norms.

Versioned checkpoint files contain wavefunctions, intrinsic Fortran RNG state, Langevin auxiliary variables, stochastic-potential state, time, step, and output index. Each checkpoint is written to a temporary HDF5 file and installed by an atomic rename. Deterministic FFTW plans permit bitwise comparison between uninterrupted and restarted runs.

6 Output and analysis

HDF5 stores physical coordinates, times, wavefunctions, populations, position expectations, and potential surfaces. Static potential surfaces are stored only in the first frame; stochastic surfaces remain frame-resolved. HDF5 operations fail loudly rather than silently leaving incomplete files. ASCII mode writes snapshot densities, observable time series, and static or frame-resolved PES files.

`ShowDyn.py` reads trajectory files, displays HDF5 trees, plots snapshots, animates densities and surfaces,

and plots reactant/product probabilities. `analyze_ensemble.py` combines rank/trajectory files or reads the parallel single-file layout, checks common time grids, writes mean and standard-error populations, performs bounded mono- and biexponential fits, and can bootstrap trajectories for empirical rate intervals.

`convergence_campaign.py` generates and executes time-step, grid, box, absorber, and ensemble-size variants, then writes CSV and Markdown acceptance tables. `mechanism_attribution.py` runs coupling-off, sub-barrier H, sub-barrier D, and over-barrier controls. It combines classical Wigner above-barrier fractions, prompt transfer, isotope suppression, background, bounded rates, and SEM. A mechanism label is emitted only when all preregistered tests pass; otherwise the result is explicitly inconclusive.

`heom_convergence.py` runs hierarchy-depth, Matsubara-pole, and retained system-basis variants. It gates population changes together with trace, Hermiticity, and positivity errors.

7 Verification status

Regression coverage includes:

- physical-unit round trips for all supported dimensions;
- zero-damping norm conservation;
- analytic white and colored centroid relaxation;
- fluctuation-dissipation ensemble means and variances;
- product-only damping isolation;
- stationary stochastic-potential initialization;
- coupling and anharmonic-potential switches;
- even/odd FFT frequency ordering and FFTW plan creation;
- smooth absorber norm removal;
- safe command-line help and explicit input paths;
- exact $t = 0$, ASCII, dynamic-PES, and HDF5 output;
- bitwise-identical checkpoint/restart with colored baths;
- two-rank single-file Parallel HDF5 MPI-IO output;
- KMS rate ratios, Gibbs stationarity, density-matrix trace, Hermiticity, and positivity;
- HEOM unitary limit, bath-correlation sum rule, active memory ADOs, strong-coupling invariants, and hierarchy-free departure;
- HEOM hierarchy-depth, Matsubara-pole, and system-basis convergence;
- two-coordinate split-operator norm conservation;
- stationary correlated HEOM residual, nonzero ADO correlations, and correlated-preparation propagation;
- exact finite spin-bath trace, positivity, and information backflow;
- three-coordinate norm conservation and external-PES round trip;
- end-to-end convergence and mechanism-control campaigns;
- HDF5 and non-HDF5 builds under strict compiler warnings.

8 Example two-timescale kinetics

A 1.5 ps product-damped trajectory at 10 K produced a bounded biexponential product-population fit with representative rates

$$k_{\text{fast}} \simeq 6.22 \times 10^{-2} \text{ fs}^{-1}, \quad (16)$$

$$k_{\text{slow}} \simeq 2.47 \times 10^{-3} \text{ fs}^{-1}. \quad (17)$$

The fit strongly improved AIC relative to a monoexponential model. Fast and slow components are consistent with prompt/over-barrier and residual tunneling channels, respectively, but this assignment is

not unique from one stochastic pure-state trajectory. Publication claims require ensemble statistics and mechanism-control calculations.

A follow-up five-control campaign used four trajectories per control, 1.5 ps duration, and two-SEM decision margins. Coupling-off background was zero and the over-barrier gates passed. The tunneling gates failed: sub-barrier transfer did not exceed estimated classical contamination by the required factor, and D transfer was not suppressed relative to H. Therefore this parameter set supports an over-barrier channel, while the slow biexponential component must remain an empirical timescale rather than a demonstrated tunneling rate.

9 Completed limitations and method boundaries

Item	Status	Note
Missing $t = 0$ output	Implemented	Exact initial frame now optional and enabled by default.
Periodic boundary return	Implemented	Smooth absorbing layers available; convergence remains user responsibility.
MPI/rank aggregation	Implemented	Ensemble tool reports means, SEM, fits, and bootstrap intervals.
Repeated static HDF5 PES	Implemented	Static PES stored once; stochastic PES remains time resolved.
Single-trajectory rate uncertainty	Implemented	Ensemble bootstrap supported when multiple trajectories exist.
Checkpoint/restart	Implemented	Versioned HDF5 stores all dynamic and random state; atomic replacement and bitwise continuation tested.
Single-file parallel HDF5	Implemented	Parallel-HDF5 build writes one fixed-shape MPI-IO ensemble file.
Quantum FDT	Implemented	Density solver enforces frequency-resolved KMS detailed balance and Gibbs stationarity.
Density-matrix decoherence	Implemented	Complete finite-grid reduced density matrix evolves under a CP Davies generator.
Multidimensional transfer	Implemented	Two- and three-coordinate, two-surface FFT propagation implemented and norm tested.
Automated convergence campaign	Implemented	Time-step, grid, box, absorber, and trajectory-count variants run automatically.
Controlled mechanism assignment	Implemented	Coupling, energy, isotope, prompt-transfer, rate, and SEM gates implemented.
Strong-coupling non-Markovian quantum bath	Implemented	Scaled Drude–Lorentz HEOM retains finite bath memory and system–bath correlations.
HEOM truncation control	Implemented	Hierarchy depth, Matsubara poles, and system-state basis are automatically converged.
Non-Gaussian bath	Implemented	Exact finite spin-bath dynamics implemented; arbitrary macroscopic non-Gaussian baths remain model specific.
Correlated thermal initial state	Implemented	Sparse stationary HEOM hierarchy and correlated system preparations implemented.
Higher than two nuclear coordinates	Implemented	Three explicit nuclear coordinates propagate with a norm-tested FFT solver.
First-principles PES interface	Implemented	Unit-checked HDF5 ingestion and interpolation implemented; electronic-structure generation remains external.

“Definitive” mechanism assignment means definitive only for the declared Hamiltonian, bath model, convergence tolerance, and controls. It is not model-independent experimental proof.

10 Recommended production protocol

1. Run `make test` for the selected compiler and library stack.
2. Perform time-step, grid-spacing, box-size, and absorber convergence.
3. Use several independent trajectories and aggregate with `analyze_ensemble.py`.
4. Monitor total norm, absorbed norm, edge density, and high-frequency FFT content.
5. Compare no-damping, reactant-only, product-only, and both-state damping.
6. Establish mechanism with barrier-energy and coupling controls before labeling fitted rates as over-barrier or tunneling.
7. Archive input, executable revision, dependency versions, seeds, raw data, and analysis outputs.

11 Build and run summary

```
make
make test
make test-parallel
./qle_1d INPUT.nml
python analyze_ensemble.py 'run.traj*.rank*.h5' --output ensemble --bootstrap 500
python ShowDyn.py run.traj000001.rank0.h5 --populations --no-usetex
python quantum_fdt_density.py --output quantum_density.h5
python heom_nonmarkovian.py --output heom_nonmarkovian.h5
python heom_nonmarkovian.py --initial-preparation correlated-projected
python heom_convergence.py --output heom_convergence
python multidimensional_transfer.py --output transfer_2d.h5
python multidimensional_transfer_3d.py --output transfer_3d.h5
python spin_bath_non_gaussian.py --output spin_bath.h5
python convergence_campaign.py INPUT.nml --executable ./qle_1d
python mechanism_attribution.py INPUT.nml --executable ./qle_1d
```

HDF5 may be disabled with `make USE_HDF5=0`. FFTW, MPI, OpenMP, and HDF5 prefixes remain configurable through the build environment. True single-file MPI-IO uses `USE_PARALLEL_HDF5=1` with a Parallel-HDF5 installation.